

CSE260

Assignment 02

This assignment must be hand-written. Show ALL steps in ALL questions.

Marks: 5+5+5 = 15

Graded Questions [1,2,3]:

1. Simplify the following boolean expressions to minimum number of literals:

$$(A \oplus B)(C+D)(A \odot C) + (A+B)(C'+D)(A'+CC')$$

$$\begin{aligned} F(A, B, C, D) &= (A \oplus B)(C+D)(A \odot C) + (A+B)(C'+D)(A'+CC') \\ &= (AB' + A'B)(C+D)(AC + A'C') + (A+B)(C'+D)A' \\ &= (AB' + A'B)(ACC + ACD + A'C'C + A'C'D) + (AA' + A'B)(C'+D) \\ &= (AB' + A'B)(AC + ACD + A'C'D) + A'BC' + A'BD \\ &= (AB' + A'B)(AC(1+D) + A'C'D) + A'BC' + A'BD \\ &= (AB' + A'B)(AC + A'C'D) + A'BC' + A'BD \\ &= AAB'C + AB'A'C'D + A'BA'C + A'BA'C'D + A'BC' + A'BD \\ &= AB'C + 0 + 0 + A'BC'D + A'BC' + A'BD \\ &= AB'C + A'BD(C'+1) + A'BC' \\ &= AB'C + A'BC' + A'BD \end{aligned}$$

2. Find the complement of the following expression:

$$\begin{aligned} &(x' + y + z')(x + y')(x' + z) \\ &= (x'.y.z') + (x.y') + (x'.z) \\ &= (x.y'.z) + (x'.y) + (x.z') \end{aligned}$$

3. Find out SOP and POS for the following [You cannot use truth table to solve these]:

$$F(V, W, X, Y, Z) = WY + WX + X'Y$$

You must show procedure for both. You must use distributive formula for POS.

$$\Sigma(2, 3, 10, 11, 12, 13, 14, 15, 18, 19, 26, 27, 28, 29, 30, 31)$$

$$\Pi(0, 1, 4, 5, 6, 7, 8, 9, 16, 17, 20, 21, 22, 23, 24, 25)$$

SOP

$$F(V,W,X,Y,Z)=WY+WX+X'Y$$

$$= WY (V+V')(X+X')(Z+Z') + WX (V+V')(Y+Y')(Z+Z') + X'Y (V+V')(W+W')(Z+Z')$$

$$= WY (VX+VX'+V'X+V'X')(Z+Z') + WX (VY+VY'+V'Y+V'Y')(Z+Z') + X'Y (VW+VW'+V'W+V'W')(Z+Z')$$

$$= WY (VXZ+VXZ'+VX'Z+VX'Z'+V'XZ+V'XZ'+V'X'Z+V'X'Z') + WX (VYZ+VYZ'+VY'Z+VY'Z'+V'YZ+V'YZ'+V'Y'Z+V'Y'Z') + X'Y(VWZ+VWZ'+VW'Z+VW'Z'+V'WZ+V'WZ'+V'W'Z+V'W'Z')$$

$$= WYVXZ + WYVXZ' + WYVX'Z + WYVX'Z' + WYV'XZ + WYV'XZ' + WYV'X'Z + WYV'X'Z' + WXVYZ + WXVYZ' + WXVY'Z + WXVY'Z' + WXV'YZ + WXV'YZ' + WXV'Y'Z + WXV'Y'Z' + X'YVWZ + X'YVWZ' + X'YVW'Z + X'YVW'Z' + X'YV'WZ + X'YV'WZ' + X'YV'W'Z + X'YV'W'Z'$$

$$= VWXYZ + VWXYZ' + VWX'YZ + VWX'YZ' + V'WXYZ + V'WXYZ' + V'WX'YZ + V'WX'YZ' + VWXYZ + VWXYZ' + VWXY'Z + VWXY'Z' + V'WXYZ + V'WXYZ' + V'WX'YZ + V'WX'YZ' + VWX'YZ + VWX'YZ' + VW'X'YZ + VW'X'YZ' + V'WX'YZ + V'WX'YZ' + V'W'X'YZ + V'W'X'YZ'$$

$$= \sum(11111, 11110, 11011, 11010, 01111, 01110, 01011, 01010, 11111, 11110, 11101, 11100, 01111, 01110, 01101, 01100, 11011, 11010, 10011, 10010, 01011, 01010, 00011, 00010)$$

$$= \sum(31, 30, 27, 26, 15, 14, 11, 10, 31, 30, 29, 28, 15, 14, 13, 12, 27, 26, 19, 18, 11, 10, 3, 2)$$

$$= \sum(2, 3, 10, 11, 12, 13, 14, 15, 18, 19, 26, 27, 28, 29, 30, 31)$$

POS

$$F(V,W,X,Y,Z)=WY+WX+X'Y$$

$$= (WY+WX+X')(WY+WX+Y)$$

$$= (WY+(W+X')(X+X')) (WY+(W+Y)(X+Y))$$

$$= (WY+W+X') (WY+(W+Y)(X+Y))$$

$$= (W+X'+W)(W+X'+Y)((W+Y)(X+Y)+W)(W+Y)(X+Y)+Y)$$

$$= (W+X')(W+X'+Y)(W+Y)(W+X+Y)(X+Y)$$

$$= (W+X'+VV')(W+X'+Y+VV')(W+Y+VV')(W+X+Y+VV')(X+Y+VV')$$

$$= (V+W+X')(V'+W+X')(V+W+X'+Y)(V'+W+X'+Y)(V+W+Y)(V'+W+Y)(V+W+X+Y)(V'+W+X+Y)(V+X+Y)(V'+X+Y)$$

$$=$$

$$(V+W+X'+Y)(V+W+X'+Y')(V'+W+X'+Y)(V'+W+X'+Y')(V+W+X'+Y+Z)(V+W+X'+Y+Z')$$

$$(V'+W+X'+Y+Z)(V'+W+X'+Y+Z')(V+W+X+Y)(V+W+X'+Y)(V'+W+X+Y)(V'+W+X'+Y)($$

$$V+W+X+Y+Z)(V+W+X+Y+Z')(V'+W+X+Y+Z)(V'+W+X+Y+Z')(V+W+X+Y)(V+W'+X+Y)$$

$$(V'+W+X+Y)(V'+W'+X+Y)$$

$$=$$

$$(V+W+X'+Y+Z)(V+W+X'+Y+Z')(V+W+X'+Y'+Z)(V+W+X'+Y'+Z')(V'+W+X'+Y+Z)(V'+$$

$$W+X'+Y+Z')(V'+W+X'+Y'+Z)(V'+W+X'+Y'+Z')(V+W+X'+Y+Z)(V+W+X'+Y+Z')(V'+W$$

$$+X'+Y+Z)(V'+W+X'+Y+Z')(V+W+X+Y+Z)(V+W+X+Y+Z')(V+W+X'+Y+Z)(V+W+X'+Y+$$

$$Z')(V'+W+X+Y+Z)(V'+W+X+Y+Z')(V'+W+X'+Y+Z)(V'+W+X'+Y+Z')(V+W+X+Y+Z)(V+$$

$$W+X+Y+Z')(V'+W+X+Y+Z)(V'+W+X+Y+Z')(V+W+X+Y+Z)(V+W+X+Y+Z')(V+W'+X+Y$$

$$+Z)(V+W'+X+Y+Z')(V'+W+X+Y+Z)(V'+W+X+Y+Z')(V'+W'+X+Y+Z)(V'+W'+X+Y+Z')$$

$$= \Pi (00100, 00101, 00110, 00111,$$

$$10100, 10101, 10110, 10111, 00100, 00101, 10100, 10101, 00000, 00001, 00100, 00$$

$$101, 10000, 10001, 10100, 10101, 00000, 00001, 10000, 10001, 00000, 00001, 0100$$

$$0, 01001, 10000, 10001, 11000, 11001)$$

$$= \Pi$$

$$(4, 5, 6, 7, 20, 21, 22, 23, 4, 5, 20, 21, 0, 1, 16, 17, 1, 0, 4, 5, 16, 17, 20, 21, 0, 1, 8, 9, 16, 17, 24$$

$$, 25)$$

$$= \Pi (0, 1, 4, 5, 6, 7, 8, 9, 16, 17, 20, 21, 22, 23, 24, 25)$$

Ungraded Questions [4, 5, 6, 7]:

4. Simplify the following boolean expressions to minimum number of literals:

$$(A'B+B')BCC'+CB'C+D$$

$$= A'BBCC' + B'BCC' + CB'C + D$$

$$= 0 + 0 + B'C + D$$

$$= B'C + D$$

5. Draw the following expression using NAND gate only. You cannot use more than 2 inputs in the gates: $F(A,B,C,D) = AB + C'D$

Do it yourself.

6. Draw the following expression using NAND gate only. You cannot use more than 2 inputs in the gates: $F(A,B,C,D) = AB + C'D$

Do it yourself.

7. Find out SOP and POS for the following [You cannot use truth table to solve these]:

$$F(A,B,C,D,E) = A + B'CD'$$

SOP

$$\begin{aligned} F(A,B,C,D,E) &= A + B'CD' \\ &= A(B+B')(C+C')(D+D')(E+E') + B'CD'(A+A')(E+E') \\ &= (AB + AB')(CD+CD'+C'D+C'D')(E+E') + B'CD'(AE+AE'+A'E+A'E') \\ &= (AB + AB')(CDE+CD'E+C'DE+C'D'E + CDE'+CD'E'+C'DE'+C'D'E') + AEB'CD' + AE'B'CD' + A'EB'CD' + A'E'B'CD' \\ &= ABCDE+ABCD'E+ABC'DE+ABC'D'E + ABCDE'+ABCD'E'+ABC'DE'+ABC'D'E' + AB'CDE+AB'CD'E+AB'C'DE+AB'C'D'E + AB'CDE'+AB'CD'E'+AB'C'DE'+AB'C'D'E' + AB'CD'E+AB'CD'E'+A'B'CD'E+A'B'CD'E' \\ &= \sum(11111, 11101, 11011, 11001, 11110, 11100, 11010, 11000, 10111, 10101, 10011, 10001, 10110, 10100, 10010, 10000, 10101, 10100, 00101, 00100) \\ &= \sum(00100, 00101, 10000, 10001, 10010, 10011, 10100, 10101, 10110, 10111, 11000, 11001, 11010, 11011, 11100, 11101, 11110, 11111) \\ &= \sum(4, 5, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31) \end{aligned}$$

POS

$$\begin{aligned} F(A,B,C,D,E) &= A + B'CD' \\ &= (A+B')(A+CD') \\ &= (A+B')(A+C)(A+D') \\ &= (A+B'+CC')(A+C+BB')(A+D'+BB') \\ &= (A+B'+C)(A+B'+C')(A+C+B)(A+C+B')(A+D'+B)(A+D'+B') \\ &= (A+B'+C+DD')(A+B'+C'+DD')(A+C+B+DD')(A+C+B'+DD') \\ &\quad (A+D'+B+CC')(A+D'+B'+CC') \\ &= (A+B'+C+D)(A+B'+C+D')(A+B'+C'+D)(A+B'+C'+D') \\ &\quad (A+C+B+D)(A+C+B+D')(A+C+B'+D)(A+C+B'+D') \\ &\quad (A+D'+B+C)(A+D'+B+C')(A+D'+B'+C)(A+D'+B'+C') \\ &= (A+B'+C+D+EE')(A+B'+C+D'+EE')(A+B'+C'+D+EE')(A+B'+C'+D'+EE') \\ &\quad (A+C+B+D+EE')(A+C+B+D'+EE')(A+C+B'+D+EE')(A+C+B'+D'+EE') \\ &\quad (A+D'+B+C+EE')(A+D'+B+C'+EE')(A+D'+B'+C+EE')(A+D'+B'+C'+EE') \\ &= (A+B'+C+D+E)(A+B'+C+D+E')(A+B'+C+D'+E)(A+B'+C+D'+E')(A+B'+C'+D+E)(A+B'+C'+D+E') \\ &\quad (A+B'+C'+D'+E)(A+B'+C'+D'+E')(A+B+C+D+E)(A+B+C+D+E')(A+B+C+D'+E)(A+B+C+D'+E') \\ &\quad (A+B+C+D'+E')(A+B+C+D'+E')(A+B+C'+D+E)(A+B+C'+D+E')(A+B+C'+D'+E)(A+B+C'+D'+E') \\ &\quad (A+B+C'+D'+E')(A+B+C'+D'+E')(A+B+C'+D'+E')(A+B+C'+D'+E') \\ &= \prod(01000, 01001, 01010, 01011, 01100, 01101, 01110, 01111, 00000, 00001, 00010, 00011, 01000, 01001, 01010, 01011, 00010, 00011, 00110, 00111, 01010, 01011, 01110, 01111) \\ &= \prod(8, 9, 10, 11, 12, 13, 14, 15, 0, 1, 2, 3, 8, 9, 10, 11, 2, 3, 6, 7, 10, 11, 14, 15) \\ &= \prod(0, 1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15) \end{aligned}$$

